

1) How to add a New Node at the begining .

Definition :

Insertion at the beginings of a linked is the process of adding a New Node as the first Node in the list . the New Node becomes the New head of the linked list . this Operation is simple and efficient because it does not require traversing.

algorithm :

- *1) Create a New node using dynamic
- *2) memory allocation .
- *3) Enter the data into the New node .
- *4) check whether the linked list is empty

Working :

A New Node containing the data '5' is created the Next pointer of the new Node is made to point to current head Node (10) the head pointer is updated to pointer to the New Node .

advantages :

- Easy to implement
- Does not require traversing the entire list
- Very fast insertion
- Suitable for stack implementation using linked lists .

Time complexity :

Best case : $O(1)$

Worst case : $O(1)$

Space complexity :

$O(1)$ excluding memory allocated for the new node.

2) To find the length of the list :

The length of a linked list is the total number of nodes present in the list. To find the length, each node is visited one by one from the beginning, each node :

Algorithm :

1) Initialize a pointer (temp) to the head node.

2) Initialize a variable.

3) Check if the linked list is empty.

4) Traverse the linked list.

5) Increment

move

6) Repeat until temp = Null

7) Display the value of count.

Working :

The hands one starts from the head node.

advantages :

Infix to postfix :

$$K + L - M * N + (O \wedge P) * W / U / V * T + Q$$

Right to left .

$$\Rightarrow Q + T * V / U / W *) P \wedge O (+ N * M - L + K$$

Infix	stack	prefix.
Q		Q
+	+	Q
T		QT
*	+	QT
V	+	QTV
/	+	QTV
U		
/		
W		
*		
)		
+		
N		
*		
M		
-		
L		
+		
K		